

```
[1]: # Q1. Check if Lionel Messi has more than 10 achievements.
# If true, print the player's name, sport, and achievements.
# Otherwise, print that the player does not have more than 10 achievements.
```



```
player_name = "Lionel Messi"
sport = "Soccer"
achievements = 7

if achievements > 10:
    print("Player Name:", player_name)
    print("Sport:", sport)
    print("Achievements:", achievements)
else:
    print(player_name, "does not have more than 10 achievements.")
```

Lionel Messi does not have more than 10 achievements.

```
[2]: # Q2. Check if a player belongs to Tennis or has exactly 20 achievements.
# If the condition is true, print a success message.
```

```
player_name = "Roger Federer"
sport = "Tennis"
achievements = 20

if sport == "Tennis" or achievements == 20:
    print("Success! The player meets the given condition.")
else:
    print("The player does not meet the condition.")
```

Success! The player meets the given condition.

```
[3]: # Q3. Check if a player has less than 10 achievements  
# and does not play Soccer.  
# Print their details if they meet the criteria.
```

```
player_name = "Usain Bolt"  
sport = "Athletics"  
achievements = 8  
  
if achievements < 10 and sport != "Soccer":  
    print("Player Name:", player_name)  
    print("Sport:", sport)  
    print("Achievements:", achievements)  
else:  
    print("The player does not meet the criteria.")
```

```
Player Name: Usain Bolt  
Sport: Athletics  
Achievements: 8
```

```
[4]: # Q4. Take an integer input from 1 to 9999 and determine:
# 1. Whether the number is even or odd.
# 2. Whether the number is a palindrome.
# 3. Whether the number is divisible by the sum of its digits.

number = int(input("Enter an integer between 1 and 9999: "))

# Check whether the number is even or odd
if number % 2 == 0:
    print("The number is even.")
else:
    print("The number is odd.")

# Check whether the number is a palindrome
if str(number) == str(number)[::-1]:
    print("The number is a palindrome.")
else:
    print("The number is not a palindrome.")

# Calculate the sum of digits
digit_sum = sum(int(digit) for digit in str(number))

# Check divisibility by the sum of digits
if number % digit_sum == 0:
    print("The number is divisible by the sum of its digits.")
else:
    print("The number is not divisible by the sum of its digits.")

Enter an integer between 1 and 9999: 1000
The number is even.
The number is not a palindrome.
The number is divisible by the sum of its digits.
```

```
[5]: # Q5. Write a for loop that prints all the elements between -5 and 5  
# using the range() function.
```

```
for number in range(-5, 6):  
    print(number)
```

```
-5  
-4  
-3  
-2  
-1  
0  
1  
2  
3  
4  
5
```

```
[6]: # Q6. Write a for loop that prints the following list.
```

```
squares = ['red', 'yellow', 'green', 'purple', 'blue']
```

```
for color in squares:  
    print(color)
```

```
red  
yellow  
green  
purple  
blue
```

```
[7]: # Q7. Use a while loop to copy the strings 'orange' from the list
# squares to the list new_squares.
# Stop and exit the loop if the value in the list is not 'orange'.
```

```
squares = ['red', 'yellow', 'green', 'purple', 'blue']
new_squares = []
```

```
i = 0
```

```
while i < len(squares):
    if squares[i] != 'orange':
        break
```

```
    new_squares.append(squares[i])
    i += 1
```

```
print("New Squares:", new_squares)
```

```
New Squares: []
```

```
[8]: # Q8. Find the animals whose names are made of 7 letters
# using a while loop and create a separate list of such animals.
```

```
Animals = ["lion", "giraffe", "gorilla", "parrots", "crocodile", "deer", "swan"]
```

```
seven_letter_animals = []
```

```
i = 0
```

```
while i < len(Animals):
    if len(Animals[i]) == 7:
        seven_letter_animals.append(Animals[i])
```

```
    i += 1
```

```
print("Animals with 7 letters:", seven_letter_animals)
```

```
Animals with 7 letters: ['giraffe', 'gorilla', 'parrots']
```